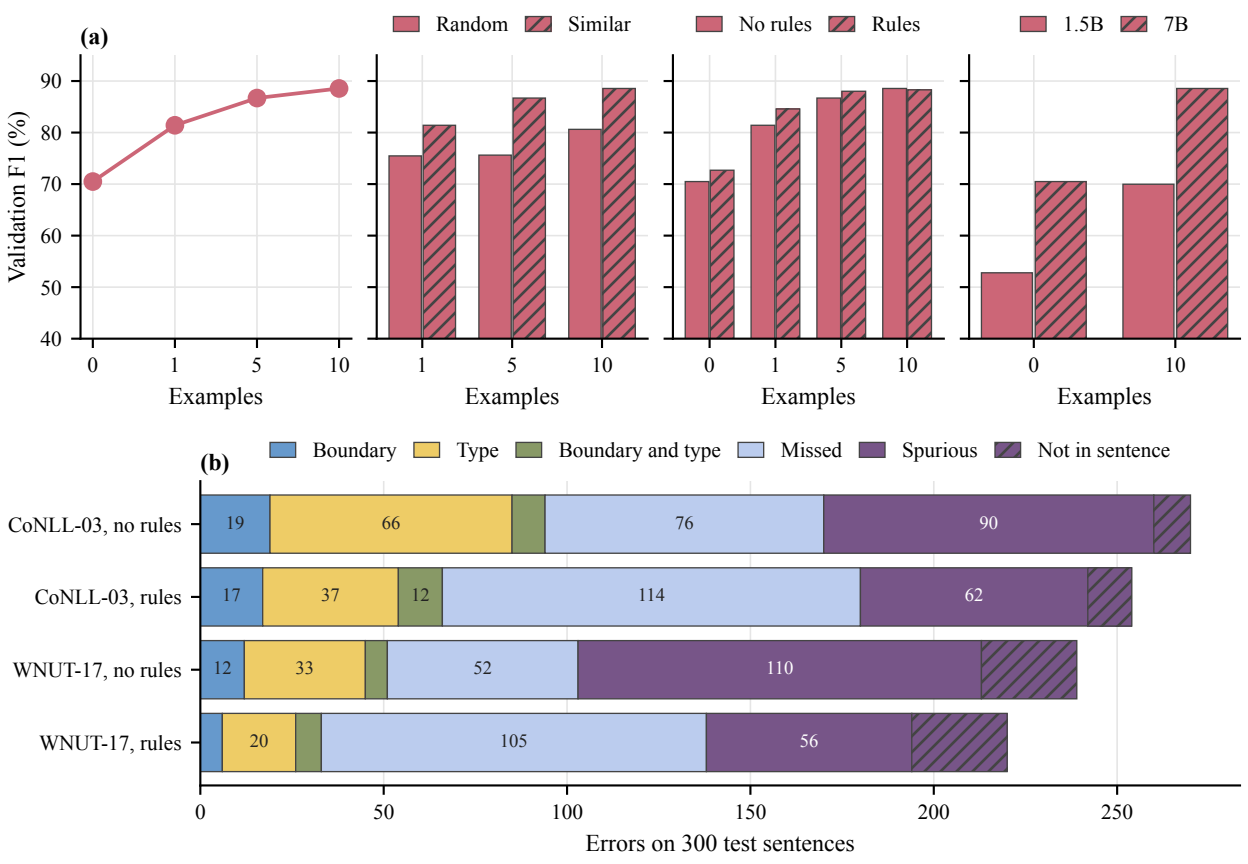
(a)
Random
Similar
No rules
Rules
1.5B
7B
Validation F1 (%)
90
80
70
60
50
40
0
1
5
10
Examples
(b)
Boundary
Type
Boundary and type
Missed
Spurious
Not in sentence
CoNLL-03, no rules
19
66
76
90
CoNLL-03, rules
17
37
12
114
62
WNUT-17, no rules
12
33
52
110
WNUT-17, rules
20
105
56
0
50
100
150
200
250
Errors on 300 test sentences